

Aletia Index — DevOps State Document

Paste this at the start of every DevOps session. Update at the end of each session.
Last updated: 2026-03-04

1. Project Identity

Field	Value
Product	Aletia Index — Clinical assurance for digital health
Owner	Annemarie Maidment (annemarie.maidment@gmail.com)
Live site	https://www.aletia-index.com
Staging	https://aletia-index.vercel.app
Repository	https://github.com/AnnMaidment/Aletia-Index
Hosting	Vercel (auto-deploy from <code>main</code>)
Database	PostgreSQL via Supabase
Domain registrar	Hostinger → Vercel DNS

2. Tech Stack

Layer	Technology	Notes
Frontend	Next.js 16 + TypeScript	App router
Styling	Inline <code><style></code> per page	⚠️ Tech debt — see CSS consolidation task
Database	PostgreSQL via Supabase	6 tables
Auth	Supabase Auth	📅 Planned — not yet implemented
Payments	Stripe + Peach Payments	📅 Planned — V3
Hosting	Vercel	Auto-deploys from <code>main</code>
External API	openFDA API	lib/fda.ts + lib/fdaSync.ts

3. Branching Strategy

main

← production (auto-deploys to Vercel)

dev

← staging (test here before merging to main)

feature/xxx

← branch from dev, merge back to dev via PR

Rule: Never push directly to `main`.

4. Environment Variables

Variable	Purpose	Required
<code>NEXT_PUBLIC_SUPABASE_URL</code>	Supabase project URL	Yes
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Supabase public key	Yes
<code>OPENFDA_API_KEY</code>	FDA API authentication	Recommended
<code>SYNC_SECRET</code>	Secures <code>/api/fda-sync</code> endpoint	Yes (production)

⚠ RLS Note: `NEXT_PUBLIC_SUPABASE_ANON_KEY` is client-exposed (by design in Supabase). When auth goes live in V2/V3, Row Level Security (RLS) policies **must** be in place on all tables before manufacturer/auditor roles are introduced. This is a pre-auth prerequisite.

Set in Vercel: Dashboard → Project → Settings → Environment Variables

5. Database Schema (Supabase / PostgreSQL)

Tables

`manufacturers`

- `id` (UUID, PK), `name`, `hq_location`, `status` (Verified/Unverified), `payment` (Stripe customer ID)

`specialty_taxonomy`

- `specialty_name` (PK), `parent_cat` (e.g. Psychiatry → Mental Health)

`device_master` ← central table

- `device_id` (PK — FDA UDI / K number format)
- `manufacturer_link` (FK → manufacturers, nullable)

- `specialty_link` (FK → `specialty_taxonomy`, nullable)
- `mode`, `dependency`, `autonomy`, `intended_use`
- `ai_ml_type` (Locked / Continuous Learning)
- `accountability_tier` (1–5)
- `health_status` (Green / Amber / Red)
- `aletia_verified` (boolean)
- `country_of_origin`
- `last_automated_sync` (timestamp)
- `last_clinical_review` (timestamp)

`regional_registrations`

- `reg_id` (UUID, PK), `device_link` (FK), `country`, `regulatory_body`, `clearance_type`, `regulatory_expiry`
- ⚠ Alert trigger: `regulatory_expiry` approaching

`clinical_audits`

- `audit_id` (UUID, PK), `device_link` (FK)
- `evidence_review_date` — ⚠ Alert trigger: stale reviews
- `confidence_score` (1–10), `evidence_log`, 10-point checklist scores

`tech_specs`

- `device_link` (PK/FK), `api_type`, `ehr_compat`, `data_hosting`, `fhir_compatible`, `popia_compliant`

6. FDA API Integration

Files:

- `lib/fda.ts` — openFDA service layer (HTTP calls, typed results)
- `lib/fdaSync.ts` — maps FDA responses → Supabase schema
- `app/api/fda-sync/route.ts` — HTTP endpoint, secured with `SYNC_SECRET`

Key functions in `lib/fda.ts`:

- `get510kByKNumber(kNumber)`
- `getPMAByNumber(pmaNumber)`
- `getRecallsByKNumber(kNumber)`
- `getAdverseEventCount(deviceName)`
- `getClassificationByProductCode(productCode)` — rejects Class I

- `searchAIMLDevices()` — up to 1000 devices
- `searchAIMLByProductCodes()`

Sync endpoint `/api/fda-sync`:

Method	URL	Action
GET	<code>/api/fda-sync</code>	Health check
GET	<code>/api/fda-sync?bulk=true</code>	Sync all devices
POST	<code>/api/fda-sync</code>	Sync single device
PUT	<code>/api/fda-sync</code>	Seed DB from FDA

Health Status Logic (auto-derived):

-  Red — active recall present
-  Amber — adverse event count > 50 MDRs
-  Green — no signals
- Human auditors can override at any time.

Scope: Class II and III AI/ML devices only. Class I excluded.

7. Roadmap Status

V1 — Complete

- ☒ Public searchable device index
- ☒ Supabase database with real schema
- ☒ Device detail modal
- ☒ Search and filter by specialty and status
- ☒ Deployed to Vercel + custom domain
- ☒ All nav pages (Methodology, Insights, Clinicians, Request Review)
- ☒ Logo, favicon, OG cover image

V2 — In Progress

- ☒ openFDA API integration (automated sync)
- ☒ 114 real AI/ML devices seeded from FDA
- ☒ Multi-jurisdiction registration support (FDA / CE Mark / SAHPRA)
- ☒ `country_of_origin` field on `device_master`
- ☐ **CSS consolidation** — move inline styles → `app/globals.css`

- ☐ **Individual device report pages** — `/device/[id]` dynamic routes
- ☐ **About / Privacy / Terms pages**
- ☐ **Manufacturer portal** — claim and manage device profiles
- ☐ **Auth** — Supabase Auth for manufacturer and auditor roles

V3 — Future

- ☐ 10-Point Clinical Assurance Checklist audit workflow
- ☐ Stripe / Peach Payments integration
- ☐ SAHPRA-specific compliance fields
- ☐ API licensing and data watermarking
- ☐ Vercel Cron Job — nightly automated FDA sync

8. Key Architectural Decisions & Rationale

Decision	Rationale	Implications
Inline <code><style></code> per page	Rapid prototyping speed	Must consolidate before V3; updating brand colours requires touching every file
Supabase anon key client-exposed	Standard Supabase pattern	RLS policies are mandatory before auth goes live
<code>device_id</code> matches FDA format (K number / UDI)	Direct foreign key alignment with FDA data	Simplifies sync; must handle non-FDA devices (CE Mark, SAHPRA) with a consistent ID scheme
Health status auto-derived from FDA, human-overridable	Reduces manual work; preserves clinical authority	Need audit trail for overrides — consider logging who changed status and when
<code>SYNC_SECRET</code> on FDA sync endpoint	Prevents unauthenticated triggering of bulk syncs	Must rotate if ever exposed; consider moving to cron-only in V3

9. Known Tech Debt

Item	Severity	Notes
CSS duplicated across all page files	Medium	Planned consolidation into <code>globals.css</code>
No RLS policies on Supabase tables	 High (before auth)	Must implement before manufacturer/auditor auth goes live
No audit log for health status overrides	Medium	Important for clinical trust credibility

Item	Severity	Notes
No error boundary / fallback UI for FDA API failures	Medium	If openFDA is down, sync silently fails
No rate-limit handling on FDA API calls	Low-Medium	openFDA has rate limits; bulk sync could hit them
Vercel Cron Job not yet set up	Low	Currently manual sync only

10. Active Session Log

Date	Session Summary	Decisions Made	Open Actions
2026-03-04	Initial DevOps State Document created. README ingested.	Agreed on living doc handoff approach.	Ann to confirm current sprint focus for V2.

11. Next Session Checklist

When starting a new session, paste this document and tell the DevOps brain:

- 1. What you worked on since last session
- 2. What you want to focus on this session
- 3. Any new decisions, blockers, or changes to the stack